CHAPTER 15 · DERIVATIONS

ELECTRIC CURRENT

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CURRENT AND DRIFT VELOCITY

SYLLABUS 15.2(B)

RESULT $I = Anev$

1. Wire of cross-sectional area A, n free electrons per unit volume, drift velocity v.
2. In time t the electrons move a distance vt, so all electrons in the volume Avt pass a cross-section.
3. Number = nAvt; charge Q = nAvte.
4. $\mathbf{I = \frac{Q}{t} = Anev}$.

2 OHM'S LAW AND J = ΣE

SYLLABUS 15.4(E)

RESULT $J = \sigma E \Leftrightarrow V = IR$

1. Uniform conductor of length l and area A with p.d. V: $J = \frac{I}{A}$ and $E = \frac{V}{l}$.
2. $J = \sigma E$ gives $\frac{I}{A} = \frac{\sigma V}{l}$.
3. $V = I\frac{l}{\sigma A} = I\frac{\rho l}{A} = IR$.
4. For a given conductor at constant temperature R is constant, so $V \propto I$: Ohm's law.

3 CONDUCTIVITY

SYLLABUS 15.4(F)

RESULT $\sigma = \frac{ne^2t}{m}$

1. In a field E an electron has acceleration $a = \frac{eE}{m}$.
2. Starting from zero average velocity after each collision, the mean time between collisions is t, so the drift velocity $v = at = \frac{eEt}{m}$.
3. $J = nev = \frac{ne^2t}{m}E$.
4. Comparing with $J = \sigma E$: $\mathbf{\sigma = \frac{ne^2t}{m}}$.

Use $\sigma = \frac{ne^2t}{m}$ to explain: a conductor conducts better than a semiconductor (much larger n); for a conductor, a higher temperature decreases t, so σ decreases; for a semiconductor, a higher temperature increases n, so σ increases.

4 RESISTORS IN SERIES AND IN PARALLEL

SYLLABUS 15.4

RESULT series: $R = R_1 + R_2 + \ldots$; parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

1. **Series**: the same current I flows through each resistor, but the potential differences are different.
2. Conservation of energy: $V = V_1 + V_2 + V_3$, so $IR = IR_1 + IR_2 + IR_3$ and $\mathbf{R = R_1 + R_2 + R_3}$.
3. **Parallel**: the potential difference V is the same across each resistor, but the currents are different.
4. Conservation of charge: $I = I_1 + I_2 + I_3$, so $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$ and $\mathbf{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$.